

Directed paths at minimum outdegree $k \leq 7$

A unified hand reduction and Rocq-checked finite proof

MAIN RESULT

Every nonempty finite oriented graph with minimum outdegree at least k , for $1 \leq k \leq 7$, contains a simple directed path of $2k$ arcs.

PROOF TYPE

Hand proofs for $k \leq 4$; graph reductions plus finite Hamilton-path obstructions for $k = 5, 6, 7$; all public endpoints and finite certificate replays checked in Rocq.

SNAPSHOT

Working-tree source audit dated 24 August 2026. The proof sources are in `graph-theory-rocq`; this manuscript records exact module names, certificate sizes, verification commands, and content hashes.

Audit convention

Path length always means number of arcs. Every theorem includes the nonemptiness hypothesis used by the formal statement. Large finite claims are not accepted on the authority of a SAT solver: their concrete CNFs, graph semantics, RUP replays, and final empty clauses are checked by Rocq.

Contents and reading guide

Contents

Contents and reading guide	2
1 Statement, scope, and status	3
2 The common Cheng–Keevash kernel	3
2.1 Reduction to a strong outregular graph	3
2.2 Maximum path, endpoint cycle, and predecessor set	4
2.3 Averaging and the finite arithmetic interface	4
3 The hand cases $k = 1, 2, 3, 4$	5
3.1 Case $k = 1$	5
3.2 Case $k = 2$	5
3.3 Case $k = 3$: closure on a five-cycle	5
3.4 Case $k = 4$: the seven-cycle obstruction	5
4 Finite Hamilton-path lemmas	6
4.1 Two-chord rotations	6
4.2 The eight extra C_{11} rotations	6
4.3 The C_{13} reverse-three-block family	6
4.4 The certificate principle	7
4.5 Exact finite statements checked	7
5 Case $k = 5$	7
5.1 The tight eight-cycle rows	8
5.2 The nine-cycle projection	8
6 Case $k = 6$	8
6.1 The two hand rows	8
6.2 The eleven-cycle rows	9
6.3 The ten-cycle prefix row	9
7 Case $k = 7$	9
7.1 Three tight rows	10
7.2 The C_{11} prefix-three branch	10
7.3 The C_{12} branch with $a = 1$	10
7.4 The C_{12} branch with $a = 2$	11
7.5 The C_{13} branch	12
7.6 Assembly	12
8 Why the finite certificates are proofs	12
8.1 CNF semantics	12
8.2 Hinted RUP replay	12
8.3 Direction of approximation	13
9 Formal dependency map	13
9.1 The $k = 7$ five-hypothesis interface	13
9.2 Trust boundary	14
10 Reproducibility and snapshot audit	14
10.1 Joint endpoint check	14
10.2 Critical checker and certificate check	14
10.3 Auditor checklist	15
11 References	16

Recommended audit order

Read Sections 1–3 for the theorem and common kernel; Sections 4–7 for the mathematical case analysis; Section 8 for the finite-proof mechanism; and Sections 9–10 for the formal dependency map, replay inventory, and reproducibility boundary.

1. Statement, scope, and status**Theorem A (cases $k \leq 7$)**

Let D be a nonempty finite oriented graph and let $1 \leq k \leq 7$. If every vertex of D has outdegree at least k , then D contains a simple directed path with exactly $2k$ arcs. Equivalently,

$$\ell(D) \geq 2k.$$

An *oriented graph* is a loopless finite digraph with no antiparallel pair: for distinct vertices x, y , at most one of $x \rightarrow y$ and $y \rightarrow x$ is present. A directed path is vertex-simple. The theorem is the collection of the first seven instances of Conjecture 1 in Cheng and Keevash’s paper; that conjecture is the oriented-graph (directed-girth three) specialization of Thomasse’s stronger girth conjecture.

Throughout, path length counts arcs, whereas $|C|$ is the number of vertices (equivalently, arcs) of a directed cycle. We write

$$N_X^+(v) := N^+(v) \cap X, \quad d_X^+(v) := |N_X^+(v)|.$$

The notation $\text{succ}_C(S)$ denotes $\{\text{succ}_C(s) : s \in S\}$. In set expressions we harmlessly identify a cycle C with its vertex set $V(C)$. Finally, $e(X, Y)$ counts arcs directed from X to Y , and $e(X)$ counts arcs of the induced subgraph on X .

k	Mathematical endgame	Formal endpoint
1	Two-arc elementary path	instance of <code>ck_theorem4_oriented</code>
2	Cheng–Keevash kernel plus 2-vertex bound	<code>ck_conj1_delta2</code>
3	Five-cycle predecessor closure	<code>ck_conj1_delta3</code>
4	Seven-cycle gateway and arc count	<code>ck_conj1_delta4</code>
5	Four kernel rows; C_8 by hand, C_9 checked	<code>ck_conj1_delta5</code>
6	Five kernel rows; $C_9/C_{10}/C_{11}$ endgames	<code>ck_conj1_delta6</code>
7	Eleven kernel rows; C_{10} – C_{13} endgames	<code>ck_conj1_delta7</code>

The words *checked* and *formal* have a narrow meaning in this dossier. The graph-to-CNF maps, Boolean semantics, RUP-checker soundness theorem, explicit finite proof objects, and final graph-theoretic assembly are Rocq terms accepted by the kernel. Python and external SAT solvers were used to discover and compress proof traces, but they are not logical assumptions.

The manuscript does not claim a hand proof for $k = 5, 6, 7$. It gives a conventional computer-assisted proof: a short mathematical reduction to exact finite statements, followed by proof objects checked by a small formal kernel. The formulas deliberately overapproximate the relevant graph projections, so unsatisfiability is used only in the sound direction.

2. The common Cheng–Keevash kernel**2.1 Reduction to a strong outregular graph**

Suppose that D is a counterexample for an integer d . At every vertex, select exactly d outgoing arcs; this gives a spanning d -outregular subdigraph. Take a sink strongly connected component H of that selected graph. No selected arc leaves H , so every vertex retains all d selected outneighbors inside H . Thus H is nonempty, strong, and d -outregular, and every directed path of H is a directed path of D . This is the construction used by the formal theorem reduction.

$$\ell(H) \leq \ell(D), \quad d_H^+(v) = d, \quad |V(H)| \geq 2d + 1. \quad (1)$$

The order bound follows from

$$d|V(H)| = |A(H)| \leq \binom{|V(H)|}{2}.$$

The formal theorem is in theories/invariants/strong.v. No Redei theorem or tournament Hamilton-path theorem is used anywhere in the current $k \leq 7$ proof.

2.2 Maximum path, endpoint cycle, and predecessor set

Let $L = \ell(H) < 2d$. Choose a longest path

$$P = v_0 v_1 \cdots v_L$$

whose first endpoint back-neighbor occurs as early as possible. Define

$$a := \min\{i : v_L \rightarrow v_i\}, \quad C := v_a v_{a+1} \cdots v_L v_a. \quad (2)$$

The outneighbors of v_L lie on P . The maximal cycle-bound argument of Cheng–Keevash gives $a \geq 1$. At the preceding path vertex v_{a-1} , put

$$A := N^+(v_{a-1}) \cap \{v_0, \dots, v_{a-1}\}, \quad (3)$$

$$B := N^+(v_{a-1}) \cap V(C), \quad (4)$$

$$S := B_C^- = \{\text{pred}_C(b) : b \in B\}. \quad (5)$$

Thus S consists of the cycle predecessors of B . Claims 11 and 12 from the proof of Cheng–Keevash Lemma 7, together with the strengthened local formalization, give the following precise imported statement.

Kernel lemma K

For the selected path and cycle,

$$\begin{aligned} 1 \leq a, \quad a + d \leq L, \quad |C| = L + 1 - a, \\ |A| + |S| = d, \quad |A| \leq (a - 2)_+, \quad |S| \leq d, \\ v_L \in S, \quad N^+(S) \subseteq V(C). \end{aligned}$$

Here $(x)_+ = \max\{x, 0\}$, matching truncated subtraction on natural numbers. Moreover, for every $u \in S$,

$$2d - L \leq d_S^+(u), \quad |S| + d - d_S^+(u) \leq |C|. \quad (6)$$

These facts are packaged by `kernel_full` and `kernel_full_AB` in `applications/ck3/lemma7.v`.

The inequality $|A| \leq (a - 2)_+$ is elementary but important. If $a = 1$, then A is empty by looplessness. If $a \geq 2$, the vertex v_{a-1} points neither to itself nor to v_{a-2} : the latter is forbidden because $v_{a-2} \rightarrow v_{a-1}$ is a path arc. This leaves at most $a - 2$ possible prefix outneighbors.

2.3 Averaging and the finite arithmetic interface

For an oriented graph on s vertices, the number of arcs is at most $s(s - 1)/2$. Therefore some $u \in S$ satisfies

$$2d_S^+(u) \leq s - 1.$$

Set $s = |S|$ and $r = d_S^+(u)$. The kernel yields the exact arithmetic interface

$$2d - L \leq r, \quad 2r \leq s - 1, \quad s + d - r \leq L + 1 - a, \quad d - s \leq (a - 2)_+. \quad (7)$$

Here r is the chosen averaging witness; it need not equal the minimum internal outdegree. This distinction is immaterial for the tight rows, but it is the exact formal interface used by `ckpath_kernel_cases4` through `ckpath_kernel_cases7`.

The same argument proves the integral Cheng–Keevash lower bound

$$\ell(D) \geq 2d - \left\lfloor \frac{d - 1}{2} \right\rfloor. \quad (8)$$

It is formal theorem `ck_theorem4_oriented`. It gives $L \geq 5, 7, 8, 10$ for $d = 3, 4, 5, 6$, respectively. For $d = 7$, the already proved $d = 6$ case gives the sharper $L \geq 12$ needed by the eleven-row reduction.

3. The hand cases $k = 1, 2, 3, 4$

3.1 Case $k = 1$

Choose an arc $v_0 \rightarrow v_1$, and then choose $v_2 \in N^+(v_1)$. Looplessness gives $v_2 \neq v_1$. If $v_2 = v_0$, then $v_0 \rightarrow v_1$ and $v_1 \rightarrow v_0$ are antiparallel, impossible. Hence $v_0 v_1 v_2$ is a simple two-arc path. This is also the $d = 1$ instance of `ck_theorem4_oriented`.

3.2 Case $k = 2$

If $L < 4$, the witness form of Lemma 7 gives a nonempty set S of order at most two with

$$d_S^+(u) \geq 4 - L \geq 1 \quad (u \in S).$$

Every oriented graph on at most two vertices has a vertex of internal outdegree zero. This contradiction proves $L \geq 4$. The formal endpoint is `ck_conj1_delta2`.

3.3 Case $k = 3$: closure on a five-cycle

Reduce to a strong 3-outregular counterexample H with $L \leq 5$. Kernel K gives a nonempty S with $|S| \leq 3$ and $d_S^+(u) \geq 6 - L \geq 1$. The oriented arc bound forces $|S| = 3$ and $d_S^+(u) = 1$ for every $u \in S$. Consequently,

$$|C| \geq |S| + 3 - 1 = 5, \quad |C| = L + 1 - a \leq L \leq 5. \quad (9)$$

Thus $L = 5$, $a = 1$, and $|C| = 5$. Every $u \in S$ has one outneighbor inside S and all three outneighbors inside C , so it points to both vertices of $C \setminus S$. If $z \in S$ but $\text{pred}_C(z) \notin S$, then $z \rightarrow \text{pred}_C(z)$ by this complete fan-out, whereas $\text{pred}_C(z) \rightarrow z$ is a cycle arc. This antiparallel pair is impossible. Therefore S is predecessor-closed on a single directed five-cycle. Since $v_L \in S$, closure gives $S = V(C)$, contradicting $|S| = 3$. The endpoint is `ck_conj1_delta3` in `ck3_main.v`.

3.4 Case $k = 4$: the seven-cycle obstruction

The integral lower bound gives $L \geq 7$, so a counterexample has $L = 7$. Kernel K gives $|S| \leq 4$ and internal minimum outdegree at least one. If $|S| = 3$, then $|A| = 1$, whence $a \geq 3$, but

$$8 - a = |C| \geq 3 + 4 - 1 = 6$$

gives $a \leq 2$. Hence $|S| = 4$. Choosing the averaging vertex with internal outdegree one gives $|C| \geq 7$, and therefore $a = 1$, $|C| = 7$, and $N^+(S) \subseteq C$.

It remains to exclude a directed seven-cycle C containing such a four-set S . Put

$$Q = C \setminus S, \quad R = V(H) \setminus C.$$

Then $|Q| = 3$ and $|R| \geq 2$.

1. Every $x \in R$ points into C . Otherwise the final two arcs on a shortest x -to- C path, followed by six cycle arcs through all seven cycle vertices, give an eight-arc path.
2. R is independent. An arc $y \rightarrow x$ in R , followed by $x \rightarrow C$ and six cycle arcs, again gives eight arcs. Hence every $x \in R$ has all four outneighbors in C .
3. Call $q \in Q$ *active* if $q \rightarrow x$ for some $x \in R$, and put $\sigma(q) = \text{succ}_C(q)$. No $y \in R$ points to $\sigma(q)$: for $y \neq x$, splice $y \rightarrow \sigma(q)$, the six cycle arcs to q , and $q \rightarrow x$; for $y = x$, replace $q \rightarrow \sigma(q)$ by $q \rightarrow x \rightarrow \sigma(q)$, obtain an eight-cycle, and let a second outside vertex enter it.
4. Not all three $q \in Q$ are active. Otherwise every four-set $N^+(y)$, $y \in R$, equals $C \setminus \sigma(Q)$. For each $q \rightarrow x_q$, orientedness gives $x_q \not\rightarrow q$, so $q \in \sigma(Q)$. Thus $Q = \sigma(Q)$, impossible for a nonempty proper subset of a single seven-cycle.
5. At most two gateways are active, and each has already used its cycle arc, so $e(C, R) \leq 2 \cdot 3 = 6$. But

$$28 = e(H[C]) + e(C, R) \leq 21 + e(C, R),$$

hence $e(C, R) \geq 7$, a contradiction.

This is formal lemma `odd_gateway_d4_impossible`. The assembled theorem and exact-path corollary are `ck_conj1_delta4` and `ck_conj1_delta4_path`. The older order-specific $k = 4$ enumerations are not proof dependencies.

4. Finite Hamilton-path lemmas

The remaining cases reduce an ambient counterexample to an oriented graph on a fixed directed cycle, together with selected endpoints and a small cover of forbidden Hamilton-path starts. Only explicitly listed paths are used; omitted graph constraints and omitted Hamilton paths enlarge the finite model. Therefore an UNSAT proof for the model excludes every genuine graph projection.

4.1 Two-chord rotations

Fix an endpoint q on a directed h -cycle and write the ordinary cycle order ending at q as

$$w_0 = \text{succ}_C(q), w_1, \dots, w_{h-1} = q.$$

For $1 \leq i \leq j \leq h-2$, define the Hamilton ordering

$$P(q; i, j) = (w_i, \dots, w_j, w_0, \dots, w_{i-1}, w_{j+1}, \dots, w_{h-1}). \quad (10)$$

It starts at w_i , ends at q , and uses only cycle arcs and the two possible chords

$$w_j \rightarrow w_0, \quad w_{i-1} \rightarrow w_{j+1}.$$

There are $\binom{h-1}{2}$ such paths per endpoint: 28, 36, 45, 55, 66 for $h = 9, 10, 11, 12, 13$.

4.2 The eight extra C_{11} rotations

The three-active C_{11} formula needs eight additional paths per endpoint. Write $w_0, \dots, w_{10}$ in cycle order with $w_{10} = q$. For $1 \leq t \leq 8$, use

$$(w_{t+1}, \dots, w_9, w_t, w_0, \dots, w_{t-1}, w_{10}). \quad (11)$$

Besides cycle arcs, it uses exactly

$$w_9 \rightarrow w_t, \quad w_t \rightarrow w_0, \quad w_{t-1} \rightarrow w_{10}.$$

The definitions `three_chord_rotation` and `three_chord_paths` are in `ckpath_cert_base.v`. Correctness is proved by

```
three_chord_paths11_numeric,
three_chord_path11_hamilton
```

in `ckpath_rotation_paths.v`.

4.3 The C_{13} reverse-three-block family

Cut $w_0 \dots w_{12}$ into four nonempty consecutive blocks $A \mid B \mid C' \mid D'$ and read them in the order

$$C' \mid B \mid A \mid D'.$$

The resulting path ends at q and uses exactly the three joining chords

$$\text{last}(C') \rightarrow \text{first}(B), \quad \text{last}(B) \rightarrow \text{first}(A), \quad \text{last}(A) \rightarrow \text{first}(D'). \quad (12)$$

There are $\binom{12}{3} = 220$ such paths per endpoint. They are defined and counted in `ckpath_cert_k7_base.v`. Their numeric and labelled Hamilton correctness are

```
reverse_three_block_paths13_numeric,
reverse_three_block_path13_hamilton
```

in `ckpath_rotation_paths_k7.v`.

4.4 The certificate principle

Finite-proof rule

A graph bridge constructs a valuation ρ satisfying a concrete CNF F whenever the claimed bad cycle projection exists. A checked RUP trace proves that no valuation satisfies F . Therefore the graph projection cannot exist. The proof generator may be untrusted because every retained addition is rechecked and the final empty clause is a theorem consequence of F .

The CNF uses arc variables x_{ij} , set variables selecting S or active endpoints, and mark variables selecting omitted starts or bad endpoints. It fixes the directed cycle, forbids antiparallel arcs, imposes only the needed cardinality and degree rows, and adds conditional clauses of the form: if the endpoint is selected and every required chord exists, then the path start belongs to the cover. A cyclic normalization fixes one genuinely selected root at index zero.

4.5 Exact finite statements checked

The following table states the semantic content of the concrete bases. Vertices are numbered cyclically; A denotes active gateways, S the kernel set, T marked bad or low endpoints, and M an omitted-start cover. A listed path is *realized* when all of its required chord variables are true. Every base also fixes the forward cycle arcs and forbids antiparallel pairs. Constraints omitted from this table are not used by the refutation.

Base	Exact retained constraints and checked endpoint
C_9	$ A = 3, 0 \in A, M \leq 5$. Every $e \notin A$ has $d_C^+(e) = 5$. For $e \in A$, $\text{succ}_C(e)$ and every realized two-chord start ending at e lie in M . Checked: <code>c9_base_unsatisfiable</code>
C_{10}	$ S = 6; T \geq 2, 0 \in T, T \cap S = \emptyset$. Each $s \in S$ has $d_C^+(s) = 6$. No $t \in T$ admits a realized listed path starting in $\text{succ}_C(S)$. Checked: <code>c10_base_unsatisfiable</code>
$C_{11}, k = 6$	For $m = 3$ or 4 : $ A = m, 0 \in A, M \leq 5$. Inactive vertices have $d_C^+ = 6$. Active successors and all realized listed starts lie in M ; $m = 3$ also uses the eight paths of §4.2. Checked: <code>c11_m3_base_unsatisfiable; c11_m4_base_unsatisfiable</code>
$C_{11}, k = 7$	$ S = 6$ and $d_C^+(s) = 7$ on S . At least two marked endpoints, rooted at 0 and disjoint from S , block every realized two-chord path starting in $\text{succ}_C(S)$. Checked: <code>c11_k7_bad_base_unsatisfiable</code>
C_{12} cover-four	$ A = 4, 0 \in A, M \leq 5$. Inactive vertices have $d_C^+ = 7$; active vertices have $d_C^+ \leq 6$. Active successors and all realized two-chord starts lie in M . Checked: <code>c12_k7_cover4_micro_base_unsatisfiable</code>
C_{12} low-three	$ S = 7$. At least three low marks, rooted at 0 and disjoint from S . On S , $d_C^+ = 7$; every $q \notin S$ has $d_C^+(q) \leq 7$; marked vertices have $d_C^+ \leq 4$; unmarked vertices outside S have $d_C^+ \geq 5$. Marked endpoints block realized two-chord starts from $\text{succ}_C(S)$. A specialized triple is forced marked but further marks remain allowed. Checked: <code>19 instances of c12_k7_low3_orbit_instance_unsatisfiable</code>
C_{13}	For $m \in \{3, 4, 5\}$: $ A = m, 0 \in A, M \leq 6$ and $0 \in M$. Inactive vertices have $d_C^+ = 7$; active vertices have $d_C^+ \leq 6$. Active successors and all realized two-chord and reverse-three-block starts lie in M . The A -mask is a canonical orbit representative. Checked: <code>c13_k7_orbit_union3_unsatisfiable; c13_k7_orbit_union4_unsatisfiable; c13_k7_orbit_union5_unsatisfiable</code>

These are overapproximations. In particular, the C_{12} and C_{13} bases deliberately omit the aggregate active-deficit inequalities used to derive the relevant cardinality branch in the graph. Their checked unsatisfiability is therefore stronger than the graph application requires.

5. Case $k = 5$

The current proof does not use the historical order-12-through-21 Z3 sweep. The kernel and the integral lower bound leave exactly four rows.

L	s	r	a	$ C $	Elimination
8	5	2	1	8	tight C_8 , hand
9	5	1	1	9	C_9 certificate
9	5	2	1	9	C_9 certificate
9	5	2	2	8	tight C_8 , hand

5.1 The tight eight-cycle rows

All 25 outarcs of S stay in the eight-cycle. At most 10 are internal to S , and the three complementary vertices receive at most 15 from S . Equality is forced. Thus S is complete toward $C \setminus S$. A cycle arc from $C \setminus S$ into S would have its reverse in this complete bipartite orientation, so $C \setminus S$ is successor-closed. This is impossible for a nonempty proper subset of a single cycle. The formal lemma is `no_tight_cycle8_five_set`.

5.2 The nine-cycle projection

Let $R = V(H) \setminus C$. Since $|H| \geq 11$, we have $|R| \geq 2$. The shortest-entry splice shows that every vertex of R points into C and that R is independent; hence every $y \in R$ has all five outneighbors in C and omits exactly four cycle vertices. Only $Q = C \setminus S$ can point to R . The active gateway argument gives at most three active endpoints, whereas

$$e(C, R) \geq 9 \cdot 5 - \binom{9}{2} = 9, \quad (13)$$

and each active endpoint contributes at most four arcs to R . Hence exactly three endpoints are active.

For a fixed $y \in R$, the successor of each active q and the start of every realized listed Hamilton path ending at q must be omitted by y . Otherwise the Hamilton path splices with an active arc to produce a ten-arc path; when the two outside roles coincide, the second vertex of R enters the resulting ten-cycle.

C_9 finite lemma

Three active endpoints force at least six starts among the ordinary and 28 two-chord paths per endpoint; equivalently, no five-set covers all listed starts. Precisely, the checked bridge `ckpath_c9_graph_certificate_impossible` rules out a strong 5-outregular oriented H with $\ell(H) < 10$, $|H| \geq 11$, a directed C_9 , and a five-set $S \subset C$ satisfying $N^+(S) \subseteq C$.

The C_9 base has 90 named Boolean variables and 787 clauses. Its 252 listed rotations, 744 retained RUP additions in two replay chunks, and final empty clause at identifier 1530 are checked in `ckpath_cert_c9.v`. The graph bridge is `ckpath_c9_graph_certificate_impossible`. The contradiction proves `ck_conj1_delta5`.

6. Case $k = 6$

The formal lower bound gives $L \geq 10$. Five rows remain. The order bound $|H| \geq 13$ is sufficient; no Redei theorem or tournament Hamilton-path theorem is used.

L	s	r	a	$ C $	Elimination
10	6	2	1	10	even-gateway hand lemma
11	5	2	3	9	tight C_9 hand lemma
11	6	1	1	11	C_{11} certificate
11	6	2	1	11	C_{11} certificate
11	6	2	2	10	prefix splice plus C_{10}

6.1 The two hand rows

For $(L, s, r, a, |C|) = (11, 5, 2, 3, 9)$, the same equality argument as in the C_8 rows makes S complete to the four complementary vertices, contradicting successor closure.

For $(10, 6, 2, 1, 10)$, the outside set is independent and points entirely into C . At most three of the four gateways are active. Since

$$e(C, R) \geq 60 - 45 = 15$$

and each active gateway contributes at most five, equality holds: exactly three are active, $H[C]$ is a tournament, and each active vertex has only its cycle successor as an outneighbor in C . The tournament induced on the active triple already has three arcs, so all three successor arcs stay inside the triple. This makes a proper set successor-closed. The formal lemma is `no_even_gateway6_c10`.

6.2 The eleven-cycle rows

The odd-gateway count gives three or four active endpoints. Fixing y outside C gives a cover of exactly five omitted cycle vertices. In the four-active branch, take the ordinary cycle path and all 45 two-chord rotations for every active endpoint. In the three-active branch, also take the eight C_{11} rotations of §4.2 for every active endpoint. In each branch, the union of the starts of all realized listed paths over all active endpoints has size at least six. Every such start must be omitted by y , a contradiction.

The two C_{11} bases have 132 named variables. The three-active formula has 2,586 clauses, 583 listed rotations, 10,005 retained RUP additions in 21 chunks, and an empty clause at identifier 12,590; its endpoint is `c11_m3_base_unsatisfiable`. The four-active formula has 2,740 clauses, 495 rotations, 4,465 additions in nine chunks, and an empty clause at identifier 7,204; its endpoint is `c11_m4_base_unsatisfiable`. The exact graph-level wrapper `ckpath_c11_graph_certificate_impossible` assumes a strong 6-outregular H with $\ell(H) < 12$, $|H| \geq 13$, a directed C_{11} , and a six-set $S \subset C$ closed under outneighbors.

6.3 The ten-cycle prefix row

Here $a = 2$. The prefix vertices v_0, v_1 lie outside C , $v_0 \rightarrow v_1$, and v_1 points to $B = \text{succ}_C(S)$. Put $Q = C \setminus S$. The six vertices of S contribute 36 internal arcs, while $H[C]$ has at most 45, so

$$\sum_{q \in Q} d_C^+(q) \leq 9, \quad \sum_{q \in Q} d_R^+(q) \geq 15. \quad (14)$$

At least three $q \in Q$ have three or more outside outneighbors. Such a q is Hamilton-bad: if a Hamilton path of C started in B and ended at q , choose an outside outneighbor different from v_0, v_1 and splice $v_0 \rightarrow v_1 \rightarrow B$, the nine cycle-path arcs, and the final outside arc. This gives twelve arcs.

C_{10} finite lemma

Among four complementary endpoints on the directed ten-cycle, at most one can block every two-chord Hamilton path whose start lies in $\text{succ}_C(S)$, when $|S| = 6$ and the vertices of S have the required internal degree.

The base has 110 named variables, 1,228 clauses, and 360 listed rotations. It has 707 retained RUP additions in two chunks and a final empty clause at identifier 1,934. Thus at most one endpoint is bad, contradicting the hand lower bound of three. The graph-to-certificate contradiction is `ckpath_c10_graph_certificate_impossible`; this completes `ck_conj1_delta6`.

7. Case $k = 7$

Let H be a strong 7-outregular counterexample. The already proved $k = 6$ theorem gives $L \geq 12$, and $L < 14$ by assumption. The oriented arc bound gives $|H| \geq 15$. Applying the kernel inequalities leaves exactly eleven rows.

L	s	r	a	$ C $	Branch
12	7	2	1	12	C_{12} cover-four
12	7	3	1	12	C_{12} cover-four
12	7	3	2	11	tight, hand
13	5	2	4	10	tight, hand
13	6	2	3	11	C_{11} bad endpoints
13	7	1	1	13	C_{13} orbit union
13	7	2	1	13	C_{13} orbit union
13	7	2	2	12	C_{12} low-three
13	7	3	1	13	C_{13} orbit union
13	7	3	2	12	C_{12} low-three
13	7	3	3	11	tight, hand

7.1 Three tight rows

For $(s, |C|) = (5, 10)$ or $(7, 11)$, closure gives

$$7s = e(S) + e(S, Q).$$

But $e(S) \leq \binom{s}{2}$ and $e(S, Q) \leq s(|C| - s)$, and the sum of these two upper bounds is exactly $7s$: respectively,

$$10 + 25 = 35, \quad 21 + 28 = 49.$$

Equality is therefore forced in both bounds. Thus $H[S]$ is a tournament, S is complete toward $Q = C \setminus S$, and every vertex of S has internal outdegree $r = 2$ or 3 . No cycle arc can go from Q to S ; hence Q is successor-closed, impossible for a nonempty proper subset of C . This eliminates the C_{10} row and both tight C_{11} rows.

7.2 The C_{11} prefix-three branch

The remaining C_{11} row has $L = 13$, $|S| = 6$, $a = 3$. The prefix data are $v_0 \rightarrow v_1 \rightarrow v_2$, and v_2 points to every vertex of $\text{succ}_C(S)$. A cycle endpoint q outside S is *bad* if no Hamilton path starts in $\text{succ}_C(S)$ and ends at q . The internal arc count forces at least three bad endpoints: writing

$$\rho(q) := 7 - d_C^+(q),$$

one obtains

$$\sum_{q \in C \setminus S} \rho(q) \geq 22$$

over the five complementary vertices, so at least three have $\rho(q) \geq 4$. Each such endpoint has an outside outneighbor beyond the three prefix vertices; a Hamilton path would splice to fourteen arcs.

C_{11} , $k = 7$, finite lemma

On a directed eleven-cycle with a six-set S whose vertices have all seven outneighbors on the cycle, two endpoints outside S cannot both block every listed two-chord Hamilton path starting in $\text{succ}_C(S)$. Thus at most one endpoint is bad.

The concrete base has 132 variables and 2,080 clauses. Five checked RUP chunks add 2,254 retained clauses and derive the empty clause at identifier 4,333. Its graph bridge is `ckpath_k7_c11_graph_certificate_impossible`. This is already a closed dependency of the $k = 7$ assembly and is not one of its five displayed final hypotheses.

7.3 The C_{12} branch with $a = 1$

Here $L = 12$, $|C| = 12$, and $|S| = 7$. Since $|H| \geq 15$, at least three vertices lie outside C . The shortest-entry argument makes the outside set independent and gives every outside vertex all seven outneighbors in C . Let $G \subseteq C \setminus S$ be the active gateways. The internal arc count gives

$$\sum_{q \in G} (7 - d_C^+(q)) \geq 18, \tag{15}$$

so $|G| \geq 3$; the gateway-omission argument excludes all five gateways being active. If $|G| = 3$, equality gives $e(C, R) = 18$ and makes $H[C]$ a tournament. Each active gateway then has outside degree six and cycle outdegree

one. The active triple already spans the three tournament arcs, so every active vertex's unique cycle outneighbor lies in G ; hence G is successor-closed, impossible. Thus $|G| = 4$.

For any outside y , the five-set

$$M = C \setminus N^+(y)$$

contains the successor of every active q and the start of every realized two-chord Hamilton path ending at q . The same splice works whether or not y is the active witness: in the coincident case one obtains a thirteen-cycle and lets another outside vertex enter it.

C_{12} cover-four lemma

No oriented directed twelve-cycle has four rooted active gateways, inactive cycle outdegree seven, active cycle outdegree at most six, and a five-set covering every active successor and every realized two-chord start.

The base has 156 variables and 5,639 clauses. A strict-RUP core with 4,847 retained additions is replayed in 194 microblocks. The endpoint theorem is `c12_k7_cover4_micro_base_unsatisfiable`.

7.4 The C_{12} branch with $a = 2$

Now $L = 13$ and the two prefix vertices v_0, v_1 precede the twelve-cycle, with

$$N^+(v_1) = \text{succ}_C(S).$$

Put $Q = C \setminus S$ and let

$$\rho(q) := 7 - d_C^+(q) \quad (q \in Q).$$

The seven vertices of S contribute 49 internal arcs, so

$$\sum_{q \in Q} \rho(q) \geq 18. \quad (16)$$

Call q *low* when $d_C^+(q) \leq 4$, equivalently when q has at least three outside outneighbors. Every low q is bad: a Hamilton path from $\text{succ}_C(S)$ to q , followed by an outside outneighbor distinct from v_0, v_1 , would give fourteen arcs. The deficit bound first gives at least two low endpoints: nonlow endpoints have outside degree at most two, low endpoints at most six, and

$$18 \leq 2 \cdot 5 + 4|\text{Low}|. \quad (17)$$

If exactly two endpoints q, r were low, every intervening bound would be tight. In particular, $H[C]$ would be a tournament and

$$d_C^+(q) = d_C^+(r) = 1.$$

Choose the direction $q \rightarrow r$. Since q 's unique cycle outneighbor is $\text{succ}_C(q)$, we have $r = \text{succ}_C(q)$. Choose $p \in S$ distinct from $\text{pred}_C(q)$, put $x = \text{succ}_C(p)$, and write the ordinary cycle order from x to p as

$$x, \dots, z, q, r, \dots, p.$$

Replace it by

$$x, \dots, z, r, \dots, p, q.$$

Tournament totality and the degree-one rows give $z \rightarrow r$ and $p \rightarrow q$, so this is a Hamilton ordering from $x \in \text{succ}_C(S)$ to q . The constructive theorem is `c12_k7_low_arc_hamilton`; `c12_k7_low_exact_two_hamilton` selects the direction between the two low vertices.

The prefix-and-fresh-outneighbor splice now gives fourteen arcs, a contradiction. Thus `c12_k7_low_card_ge3` proves $|\text{Low}| \geq 3$. The finite model marks a chosen low triple. Rotation of the twelve-cycle reduces the $\binom{12}{3} = 220$ triples to 19 cyclic orbit representatives.

C_{12} low-three family

For each of the 19 cyclic representatives of a three-set, the specialized CNF forcing that chosen triple to be low, complementary to S , and blocking all listed starts is unsatisfiable. The formula soundly permits further endpoints to be marked low.

The unspecialized low-three base has 7,132 clauses; substitution produces the 19 concrete instances. Their checked strict-RUP replays total 316 microblocks. The formal orbit table, specialization soundness, graph valuation, and each final empty clause are all inside Rocq.

7.5 The C_{13} branch

The final three rows have $L = 13$, $a = 1$, $|C| = 13$, and $|S| = 7$. At least two vertices lie outside C . They form an independent set and each has all seven outneighbors in C . Let $G \subseteq Q = C \setminus S$ be active. The deficit count gives

$$\sum_{q \in G} (7 - d_C^+(q)) \geq 13, \quad 3 \leq |G| \leq 5. \quad (18)$$

For an outside vertex y , the six-set $M = C \setminus N^+(y)$ covers all starts of realized Hamilton paths ending at active gateways. Two-chord rotations alone are not strong enough for $|G| = 3$ or 4 ; adding the 220 reverse-three-block paths per endpoint closes the gap.

C_{13} finite lemma

Fix a rooted active gateway set G of order 3, 4, or 5 on a directed thirteen-cycle. If inactive vertices have cycle outdegree seven and active vertices have cycle outdegree at most six, no six-set M containing the root can cover all active successors and all realized ordinary, two-chord, and reverse-three-block starts.

Cyclic symmetry is handled without trusting a symmetry-breaking solver option. Rotate G to a canonical cyclic representative rooted at zero. There are 22, 55, and 99 representatives at weights 3, 4, and 5. One orbit-union formula per weight blocks every other rooted mask, so a genuine graph valuation satisfies exactly one of three bases. For the extra unit $0 \in M$, choose an outside witness y of the rooted active q . Since $q \rightarrow y$, orientedness gives $y \not\rightarrow q$; hence $q \in C \setminus N^+(y) = M$. This is `c13_k7_orbit_union_satisfied_at_root`.

$ G $	Canonical masks	Base clauses	RUP additions	Microblocks
3	22	13,943	48,767	488
4	55	14,844	9,410	95
5	99	15,933	2,281	23

All retained additions are strict RUP; no RAT step is trusted. The three final theorems are `c13_k7_orbit_union3_unsatisfiable`, `c13_k7_orbit_union4_unsatisfiable`, and `c13_k7_orbit_union5_unsatisfiable`.

7.6 Assembly

The eleven rows are now exhausted: three tight rows by successor closure, the prefix-three C_{11} row by the bad-endpoint certificate, two C_{12} , $a = 1$, rows by cover-four, two C_{12} , $a = 2$, rows by the low-three orbit family, and three C_{13} rows by the orbit-union bases. Therefore no strong 7-outregular counterexample exists. The standard reduction returns the result for the original graph. The unconditional endpoint is `ck_conj1_delta7`; `ck_conj1_delta7_path` returns an explicit 14-arc prefix of a longest path.

8. Why the finite certificates are proofs

8.1 CNF semantics

A literal is a variable identifier with a polarity; a clause is a finite list of literals; a CNF is a finite list of clauses. The definitions `eval_literal`, `satisfies_clause`, and `satisfies_cnf` are transparent Rocq functions in `ckpath_cnf.v`. Cardinality clauses are generated by transparent combinatorial functions and guarded so that graph-semantic degree equalities are imposed only where justified.

8.2 Hinted RUP replay

To add a candidate clause Γ by reverse unit propagation, the checker assumes every literal of Γ false and follows a supplied list of existing clauses. Each listed clause must be unit under the current partial assignment and therefore forces its last undecided literal. The step succeeds only if propagation reaches a contradiction. The soundness theorem proves that every valuation satisfying the current database also satisfies Γ . Repeating this argument preserves database satisfaction; a checked empty clause makes satisfaction impossible.

Deletion records are not a source of strength: removing a clause only weakens the active database, and the soundness invariant is threaded explicitly. For the large $k = 7$ traces, microblocks import only selected base clauses and selected consequences of earlier producer blocks. The explicit microblock checker verifies those selections and each local RUP sequence, while opaque theorems hide large intermediate terms after checking.

8.3 Direction of approximation

Soundness direction

Every counterexample graph induces a satisfying valuation of the finite CNF. The CNF may admit extra valuations because it omits ambient vertices, many graph constraints, and most Hamilton paths. Proving the larger set of valuations empty is stronger than necessary. No step infers graph existence from CNF satisfiability.

9. Formal dependency map

Layer	Principal modules	Role
Graph foundation	digraph, oriented, dipath, strong	finite oriented graphs, paths, SCC reduction
CK kernel	ck3/lemma7.v	maximum path, Claims 11–12, counting
Arithmetic	ckpath_shapes.v; ckpath_kernel_cases.v	exact rows for $k = 4, 5, 6, 7$
Hand endgames	ckpath_tight_cycles.v; ckpath_odd_gateway.v; ckpath_even_gateway.v	successor closure and gateway counts for $k \leq 6$
$k = 7$ hand layer	ckpath_k7_tight.v; ckpath_k7_even_gateway.v; ckpath_k7_c11_hand.v; ckpath_k7_c12_low_hand.v; ckpath_k7_c12_low_arc_hamilton.v; ckpath_k7_c12_low_ge3.v	tight rows, gateway count, bad/low endpoints, equality splice
Path semantics	ckpath_rotation_paths.v; ckpath_rotation_paths_k7.v	numeric rotations become labelled Hamilton paths
Boolean checker	ckpath_cnf.v; ckpath_drup.v; ckpath_drup_explicit_microblock.v	CNF semantics and RUP soundness
Graph bridges	ckpath_cert_graph_*.v	counterexample valuations satisfy CNFs
Assemblies	ck3_main.v; ckpath_k4.v; ckpath_k5.v; ckpath_k6.v; ckpath_k7.v; ckpath_k7_concrete.v	public theorems

k	Public theorem	Source
1	ck_theorem4_oriented ($d = 1$)	applications/ck3/lemma7.v
2	ck_conj1_delta2	applications/ck3/lemma7.v
3	ck_conj1_delta3	applications/ck3/ck3_main.v
4	ck_conj1_delta4	applications/ck_path/ckpath_k4.v
5	ck_conj1_delta5	applications/ck_path/ckpath_k5.v
6	ck_conj1_delta6	applications/ck_path/ckpath_k6.v
7	ck_conj1_delta7	applications/ck_path/ckpath_k7_concrete.v

9.1 The $k = 7$ five-hypothesis interface

The intermediate theorem `ck_conj1_delta7_from_finite_refutations` displays exactly five finite assumptions: C_{12} cover-four, the 19-instance C_{12} low-three family, and the three C_{13} orbit-union bases. The C_{11} theorem is already closed. The file `ckpath_k7_concrete.v` supplies all five checked refutations and exports the unconditional theorem. A source audit finds no `Admitted`, `admit`, `Axiom`, or `Parameter` in the selected `ck3` or `ck_path` proof sources. Section-local hypotheses are theorem parameters, not global assumptions, and are discharged at the public endpoints.

9.2 Trust boundary

- **Trusted:** the Rocq 9.1.1 kernel and toolchain used to compile and check terms; the imported MathComp, HB, Stdlib, and local foundational graph libraries; and the statement-to-intended-mathematics correspondence.
- **Not trusted:** Z3 or any other SAT solver; Python generators; proof trimming; orbit discovery; scheduling scripts; rocq-evolve transport; and this PDF generator. These may create candidate data, but the theorem does not assume their correctness.
- **Review obligation:** a formal proof can establish only its formal statement. The endpoint states nonempty finite oriented graphs, minimum outdegree at least k , and a simple directed path of $2k$ arcs. Those are the conventions used throughout this manuscript.

10. Reproducibility and snapshot audit

10.1 Joint endpoint check

During preparation, a joint check first detected stale $k = 4$ and $k = 5$ object files: they referred to older digests of `ckpath_kernel_cases` and `ckpath_cert_graph_odd`. Recompiling those two public modules against the current dependencies removed the inconsistency. Run from `graph-theory-rocq/digraph-theory`, the following joint command then exited successfully with no output:

```
opam exec --switch=rocq-tools -- rocqchk -silent \
-R theories Digraph -Q ../base/theories GTBase \
-norec Digraph.applications.ck3.ck3_main \
-norec Digraph.applications.ck_path.ckpath_k4 \
-norec Digraph.applications.ck_path.ckpath_k5 \
-norec Digraph.applications.ck_path.ckpath_k6 \
-norec Digraph.applications.ck_path.ckpath_k7_concrete
```

In `rocqchk` syntax, `-norec MODULE` checks that module while admitting its already compiled dependencies. Repeating the option is essential: writing it only once before several modules checks the later modules recursively and can trigger a very long monolithic traversal.

10.2 Critical checker and certificate check

A second joint nonrecursive check of the RUP checkers, CNF reduction, $C_9/C_{10}/C_{11}$ certificates, C_{12} orbit family, C_{13} concrete wrapper, and finite $k = 7$ assembly completed successfully. Run from the same directory, the exact command was:

```
opam exec --switch=rocq-tools -- rocqchk -silent \
-R theories Digraph -Q ../base/theories GTBase \
-norec Digraph.applications.ck_path.ckpath_drup \
-norec Digraph.applications.ck_path.ckpath_drup_explicit_microblock \
-norec Digraph.applications.ck_path.ckpath_cnf_reduction \
-norec Digraph.applications.ck_path.ckpath_cert_c9 \
-norec Digraph.applications.ck_path.ckpath_cert_c10 \
-norec Digraph.applications.ck_path.ckpath_cert_c11_m3 \
-norec Digraph.applications.ck_path.ckpath_cert_c11_m4 \
-norec Digraph.applications.ck_path.ckpath_cert_k7_c12_low3_orbits \
-norec Digraph.applications.ck_path.ckpath_cert_k7_c13_orbit_union_concrete \
-norec Digraph.applications.ck_path.ckpath_k7
```

These checks complement compilation; they are not a substitute for a full recursive audit of every imported library. The 1,267 source files under the current `ck_path` working tree have the aggregate SHA-256 digest shown below, computed from the sorted list of per-file SHA-256 lines.

Audited source-tree digest

```
34106e945de7265fad8ab6ee8fe052e2dec7a84970e4cf217215fb20ac5a520f
```

The enclosing repository HEAD was `2f32e267a637b9fc1fffb22150166f129f9a9407`, but the `ck_path` development is a dirty/untracked working-tree addition, so the aggregate digest, not the commit alone, identifies the audited snapshot.

Source	SHA-256
ck3/lemma7.v	1be940ab45f4b7a018921c779a150fa0a6c7eed31a54dfef3110ee2b5472e2c1
ck3/ck3_main.v	b2616ecc193fa84d82cf9db93d31edb87129c0cd59baaa8be585a8a4026013a8
ckpath_kernel_cases.v	5de560ab98570555bc21663fdd908501ebb7ea343b7ee6d2c59cfed447f299c9
ckpath_k4.v	78924ee8ac15eba9ae02f34ee2eccd2846add8f1d9aa3ca0f450e1f52c35206e
ckpath_k5.v	ff51da760980481b338d9f35c939dba22612f2ef311184d7e9ed8525fc16e21b
ckpath_k6.v	992bc718a049dd5d34868832c53a3261fbc48b37d273498d64d838731c900430
ckpath_k7.v	a5f740703ceb14535c683e445171d424e12ee59e154cb726b7da4722a9daa6ff
ckpath_k7_concrete.v	483bccac7d96a39fa3ec88613597fa5f59c2454d7bc2fce72643566734100c47

10.3 Auditor checklist

1. Check the reduction theorem: selected outgoing arcs stay inside the final sink SCC, and every path in the reduced graph is a path in the original graph.
2. Check Kernel K against the maximum-cycle-bound path construction, especially $a \geq 1$, $|A| \leq (a - 2)_+$, $N^+(S) \subseteq C$, and the two count inequalities in (6).
3. Recompute each displayed arithmetic row table from (7).
4. For every hand row, check that successor closure is derived from orientedness and a tight arc count, not assumed.
5. For each finite branch, verify the graph-to-CNF valuation and the direction of overapproximation.
6. Check the numeric path families and their chord lists. A path-blocking clause is sound only if the listed order is Hamiltonian and ends at the claimed endpoint.
7. Inspect the RUP-checker soundness theorem, then replay the concrete certificate modules or run the listed joint checks.
8. Use `Print Assumptions` on the public endpoints and compare the formal definitions of oriented graph, outdegree, dipath, path size, and longest-path length with Theorem A.

11. References

- [1] Yangyang Cheng and Peter Keevash, *On the Length of Directed Paths in Digraphs*, SIAM Journal on Discrete Mathematics 38 (2024), no. 4, 3134–3139. Conjecture 1, Theorem 4, Lemma 7, and Claims 11–12 are the external mathematical source. [arXiv:2402.16776](https://arxiv.org/abs/2402.16776); DOI [10.1137/24M1648375](https://doi.org/10.1137/24M1648375).

- [2] The Rocq proof sources are under the following directories in the local `graph-theory-rocq` working tree:

```
digraph-theory/theories/applications/ck3
digraph-theory/theories/applications/ck_path
```

The exact public theorem and certificate module names are listed in Sections 9 and 10.

- [3] The human-readable discovery notes and historical computational cross-checks are under

```
problems/directed_path_minimum_outdegree
```

in the `graph-conjectures` repository. Where those notes differ from the current formal proof, this dossier follows the compiled Rocq dependency graph.

End of audit manuscript. This document is explanatory; the authoritative proof objects are the source modules identified above.